School of Business, University of Kansas

# BSAN 400 Special Topics of Business Analytics (3credits)
# Introduction to Machine Learning with R
## Spring 2019

**Class Time: T, H 4 – 5:15pm, CAPF 4011**


**Instructor:** Shaobo Li, Ph.D.

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|---|---|
| Office: | CAPF 4125 |
| Office Hours: | T, H, 3 – 4pm, or by appointment |
| Email: | shaobo.li@ku.edu |
| Course web: | Blackboard |

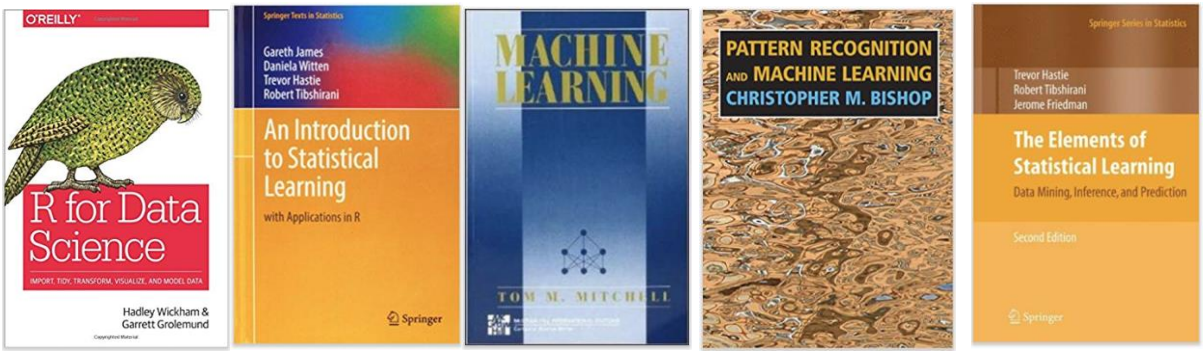

**Prerequisites:** College level courses on
- Mathematics
- Probability and statistics
- Programming

**Course description:** The first part (about 25-33%) of this course covers fundamental R programming, and second part introduces students different machine learning techniques and how they are applied in real life, and how they are implemented in R. Students will get exposure to basic as well as state-of-the-art machine techniques. The focus of this course is rather a broad view than details. The course heavily relies on programming.

**Course Outcomes:** You will be able to answer these questions generally and specifically:
- What is machine learning, and what are popular learning algorithms?
- What are principles of machine learning?
- How does a specific algorithm work?
- Why does it work?
- When does it work and when does it not work?

**Course Format:** Classes will be provided in mixed forms of lecture and lab.

**Recommended Textbook:**



**Grading:**

- Assignment 40%
  About 8 individual assignments. Later submission is NOT acceptable.
- Midterm 30%
  It consists of two parts: in-class (close book) and take home.
- Final (Take home) 20%
- In-class quiz (open book) 10%
- Discussion board (extra credit) 5%
  You are encouraged to participate discussion board to ask and answer questions.

**Academic Integrity:** The University of Kansas School of Business requires that all students conduct themselves in an ethical manner as outlined in the School of Business Honor Code. This dedication to ethical behavior is an integral part of the student experience, and academic integrity will be taken very seriously in this class. A breach of the honor code could potentially result in punitive punishments including expulsion from the University. More information about these expectations is available at: http://business.ku.edu/honor-code. If cheating is found, anyone involved will be given 0 grade, and case will be reported to university.

**Class Communication:**
We will use Blackboard to communicate. To ask questions, you may go to discussion board first to see if the same question has already been asked by someone, and if the answer is there. If not, you can send me email. To make sure that your message is not accidentally deleted as junk, please include '**BSAN400**' in the **email subject** line.

**Other policies:** You need to send me early notice if you would miss a class or have possible late submission due to health-related issue, travel, or other emergencies. The instructor reserves the right to change the syllabus.

**Handling depression**
KU Counseling and Psychological Services (CAPS)

**Tentative Schedule**

| **Week** | **Lecture and lab** | **Assignment Due** |
|---|---|---|
| Week 1 | Introduction | |
| Week 2-5 | Foundations of R programming | Assignment 1 & 2 (2/8, 2/22) |
| Week 6 | Principles and concepts | |
| Week 7 | Review and Midterm | Assignment 3 (3/8) |
| **Week 8** | **Spring break** | Take home midterm (3/17) |
| Week 9 | Decision tree (CART) | |
| Week 10 | Bagging, random forest, boosting | Assignment 4 (3/29) |
| Week 11 | Support vector machine | Assignment 5 (4/5) |
| Week 12-13 | Neural networks, Deep learning | Assignment 6 (4/12) |
| Week 14 | K-means clustering | Assignment 7 (4/26) |
| Week 15-16 | Text analytics | Assignment 8 (5/3) |
| **Final week** | **Take home exam due on 5/17 4pm.** | |